

Bhargav Krishna Yadav Challa

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Education

National Institute of Technology, Nagpur, India

Jul 2020 – May 2024

Bachelor of Technology, Electrical and Electronics Engineering

Optional Subjects: Principles of Economics, Psychology, Human Resource Management

Work Experience

Quantum Algorithm Engineer, QuantumIntegrated Machines

July 2026 – Present

- Design hybrid quantum-classical algorithms for strongly correlated chemistry, targeting chemical accuracy for drug-discovery workloads.
- Build ML pipelines that accelerate electronic structure calculations and benchmark emerging quantum methods against DMRG/CCSD(T) references.
- Partner with research and commercial teams to validate pharmaceutical use cases and translate quantum capabilities into deployable solutions.

Quantum Research Intern, QuantumIntegrated Machines

Mar 2026 – June 2026

- Implemented classical-quantum hybrid algorithms (VQE, SQD) on the algorithms side using the Qiskit framework, running end-to-end molecular ground-state energy simulations (FMO-VQE [GitHub], shadow Tomography [GitHub]).
- Profiled circuit depth, gate counts, and error budgets across ansatz choices to assess NISQ-hardware feasibility.
- Analyzed quantum hardware limits, noise, and error-correction techniques alongside the field's near-term scope.
- Assessed the current state of the industry and practical, near-term use cases in drug discovery.

Associate – Risk Consulting, PwC India

Oct 2024 – Jan 2026

- Analyzed **60+ security alerts daily** in a 24x7 Security Operations Center (SOC) using SIEM and SOAR tools (Chronicle, ArcSight), improving incident response time by **30%** with **99% SLA adherence**.
- Performed log analysis across **20+ critical assets**, delivering client-facing remediation reports that cut resolution time by **30%** and ensured **100% monitoring continuity**.

Leadership Experience

Draftsperson, Student Governance Reform Initiative, NIT Nagpur

Jan 2024 – May 2024

- Authored a **20+ page Student Constitution and Election Commission framework**, presented to 3 college deans to establish the foundation for campus-wide governance reform.
- Translated stakeholder interviews into structured policy recommendations and proposed 3 new council roles, reshaping accountability and decision-making for **3,500+ students**.

Co-Founder, VNIT-TSS Committee, NIT Nagpur

Aug 2023 – May 2024

- Founded and scaled a cultural body representing **800+ Telugu students**, driving inclusive participation across campus life.
- Directed 3 large-scale events (Sankranti & Ugadi '23-'24) on a self-funded budget, drawing **1,200+ attendees** per event.

Student Mentor, NIT Nagpur

Aug 2022 – May 2024

- Mentored **15 first-year students** over 2 years under the Student Mentorship Program (SMP), guiding academic performance, wellbeing, and campus integration.
- Managed high-impact cases including mental health crisis and **suicide prevention**, coordinating with faculty and counselors to ensure timely intervention and student safety.

Founder, CTD (Code, Think, Develop) Tech Community, NIT Nagpur

Aug 2022 – May 2023

- Founded a student tech community running peer-led project work and mentorship over its first **10 months**.

Analytical & Professional Skills

- **Programming** : Python, Machine Learning
- **Quantum Stack**: Qiskit, Qiskit Nature, VQE, QPE, QAOA, Quantum Error Correction (Basics)
- **Professional**: Communication Strategy (PwC), Cross-Functional Work, Use-Case Validation

Additional Information

- Volunteered with Prayaas, the college's social service club.
- Developed an interest in reading, exploring philosophy, nonfiction, and fiction.
- Passionate about world cinema, exploring storytelling styles across different cultures and genres.
- Deep appreciation for Indian classical music and dance — their rhythm, history, and expressive depth.